PAP Manydaughters

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H1: Having daughters leads to stronger support for gender equality (as measured by attitudes).

Variables:

Outcomes: we will consider the following items (asked in 2018):

- Agree / disagree: "Children below the age of 6 suffer if their mother works." The original item is taken from `pl/plh0298_v2`, which is measured on a 1-7 likert scale. Negative values on the original item (denoting various types of non-response) will be recoded as missing. We also rescale the original item such that higher values denote increased support for gender equality.
- Agree / disagree: "Children below the age of 3 suffer if their mother works." The original item is taken from `pl/plh0302_v2`, which is measured on a 1-7 likert scale. Negative values on the original item (denoting various types of non-response) will be recoded as missing. We also rescale the original item such that higher values denote increased support for gender equality.
- Agree / disagree: "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally." The original item is taken from `pl/plh0308_v2`, which is measured on a 1-7 likert scale. Negative values on the original item (denoting various types of non-response) will be recoded as missing. The original item is already scaled such that higher values denote increased support for gender equality.

We combine these items into a single outcome variable I using the following procedure:

Let $k = 1, ..., 3$ denote the outcomes of interest, and μ_k and σ_k denote k's average and standard deviation in the control group (no daughters).

I is then given by $I = \frac{1}{3} \sum \frac{k-\mu_k}{\sigma_k}$. In other words, I is the average of the standardized outcomes.

I is coded as missing in case of non-response on any one of the 3 sub-items. Thus, participants must provide answers to all 3 subitems in order to be included in the analysis.

Treatment: `d_daughter` is a dummy variable taking the value = 1 if the respondent has at least one daughter in 2018, and 0 if the respondent has only sons in 2018. `d_daughter` is coded as missing if the respondent has no children by 2018.

`d_daughter` will be coded based on information contained in `KIDGEB*` and `KIDSEX*` from the `Biobirth` dataset.

Controls: `numkids` is a categorical variable recording the total number of children in 2018. Information is taken from `KIDGEB*`.

`gebjahr` (year of birth) and `sex` will be taken directly from the `Biobirth` dataset. All negative values will be recoded as missing. We will treat `gebjahr` as a continuous variable, and `sex` as a dummy.

Inclusion/exclusion criteria:

We will drop respondents in the IAB-BAMF-SOEP Refugee sample (samples M3, M4 and M5). This information is contained in the variable `psample` from the `ppathl` dataset.

Otherwise, we anticipate no data exclusions except due to missing values on our variables listed above.

Importantly, this means that respondents without children in the year that outcomes are recorded are dropped from our analyses.

Exact analysis:

We will estimate the following model via OLS:

$$I_i = \beta_1 \, d_daughter_i + \beta_2 \, sex_i + \beta_3 \, gebjahr_i + \Gamma_i + \epsilon_i$$

where I_i represents our outcome variable, and Γ_i represents fixed effects for the number of children in 2018 (`numkids`), and ϵ_i denotes robust standard errors.

Interpretation of results:

Our quantity of interest is $\hat{\beta}_1$, which represents the estimated difference in attitudes for respondents with vs. without daughters.

H2 predicts that $\hat{\beta}_1$ is positive, indicating more supportive attitudes amongst respondents with daughters.

Our null hypothesis is thus $\beta_1 \leq 0$, and we will do one-tailed tests (i.e. the t-statistic $>$ 1.645).

H2: Having daughters leads to more left-leaning political preferences.

Variables:

Our **outcome** is a measure of left-right self-placement:

"In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?"

The original items are from `pl/plh0004` (asked in years 2005, 2009, 2014, 2019). We simply recode all negative values to missing.

`d_daughter` will be coded based on whether a respondent has at least one daughter in the year that outcomes are collected.

`numkids` will also be coded to reflect a respondent's birth history in the years that outcomes are collected.

`gebjahr` and `sex` will be coded in the same way as describe above.

Inclusion/exclusion criteria:

Same as above.

Exact analysis:

Since we now have repeated observations within respondents, we estimate the following model:

$$leftright_{it} = \beta_1 \; d_daughter_{it} + \beta_2 \; sex_{it} + \beta_3 \; gebjahr_{it} + \Gamma_{it} + \epsilon_{it}$$

with cluster standard errors within `pid`.

Interpretation of results:

Our quantity of interest is $\hat{\beta}_1$, which represents the estimated difference in left-right self-placement for respondents with vs. without daughters.

H2 predicts that $\hat{\beta}_1$ is negative, indicating more left-leaning political attitudes amongst respondents with daughters.

Our null hypothesis is thus $\beta_1 \geq 0$, and we will do one-tailed tests (i.e. $t <$ -1.645).

H3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Variables:

Conceptually, we want our **outcome** to be the **difference** in the amount of time that mothers and fathers spend on the following tasks:

"For the following activities, please indicate how many hours you spend on each one in an average working day:"

- Errands and shopping? (`pl/pli0040`)
- Housework such as washing, cooking, cleaning? (`pl/pli0043_h`)
- Looking after children? (`pl/pli0044_h`)

We combine these three items into a simple additive index: $I = pli0040 + pli0043_h + pli0044_h$, which is calculated for each individual respondent in the dataset.

As in H1, I is coded as missing in case of non-response on any one of the 3 sub-items.

Next, we create a variable $DI = |I_{father} - I_{mother}|$.

To do so, we first create male and female subsets of the data, and then merge the male subset (as "using" data) to the female subset (as "master" data) based on `parid`, keeping only matched observations ("_merge == 3"). Essentially, this should drop all respondents who are without a partner in a specific year.

We then construct DI as the simple difference described above. Note that since we are taking the absolute value, DI is bounded by 0. Additionally, DI can still be missing in case either I_{father} or I_{mother} is missing in a particular year due to item non-response.

Turning to the **treatment** variable, we will create `d_daughter` at the individual-level in the manner described for H2.

However, when merging the male and female subsets, we recognize that $d_daughter_{father}$ may not always be equivalent to $d_daughter_{mother}$.

The SOEP documentation also notes some differences (which we don't quite understand) in how birth histories are collected for men vs. women.

For simplicity, we code $d_daughter \equiv d_daughter_{mother}$ for our analysis.

The same logic applies to our handling of `numkids`.

Inclusion/exclusion criteria:

Same as above. Notably, we now exclude observations where `DI` is missing, which includes instances where individuals are unpartnered.

Exact analysis:

We will estimate the following model via OLS:

$$DI_{it} = \beta_1 \; d_daughter_{it} + \Gamma_{it} + \epsilon_{it}$$

where Γ_{it} denotes fixed effects for `numkids`, and standard errors are clustered based on **mothers'** `pid`.

Interpretation of results:

Our quantity of interest is $\widehat{\beta}_1$, which represents the estimated "mother-father gap" in hours committed to household activities for couples with vs. without daughters.

H3 predicts that the "mother-father gap" should be smaller for couples with girls. In other words, if the alternative hypothesis holds, we expect $\widehat{\beta}_1$ to be negative.

The null hypothesis is therefore: $\beta_1 \geq 0$, and we will do one-tailed test ($t < -1.645$).

H4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Variables:

Same variables as in H1.

Inclusion/exclusion criteria:

Same as in H1.

Exact analysis:

We will estimate the following interaction model via OLS:

$$I_i = \beta_1 \, d_daughter_i + \beta_2 \, male_i + \beta_3 \, daughter_i * male_i + \beta_4 \, gebjahr_i + \Gamma_i + \epsilon_i$$

Interpretation of results:

The quantity of interest is $\widehat{\beta}_3$, which captures how the "daughter effect" differs between men and women.

H4 predicts a stronger "daughter effect" for men (i.e. $\widehat{\beta}_3 > 0$).

Again, we will assess this hypothesis using a one-tailed test ($t > 1.645$).